WS08: Oscillation of a Bifilar Suspension **Time: 1 hr**

Name: .... ( )

Class: 26S **Date:** ....

In this experiment, you will investigate the oscillation of a bifilar suspension.

(a) **(i)** Suspend a metre rule horizontally from another metre rule using two vertical threads as shown in Fig. 1.

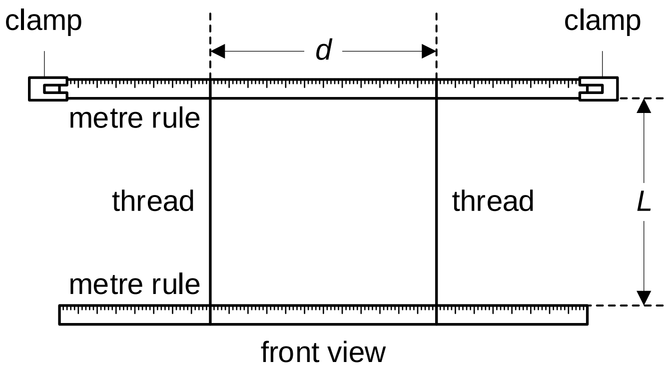


Fig. 1

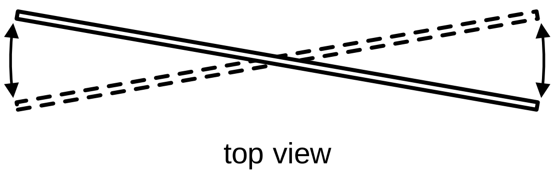


Fig. 2.

(ii) Adjust both threads to a length L of about 30 cm. Arrange the threads at a distance d of 80 cm apart and ensure that the threads are equidistant from both ends of the rules. Measure and record d.

Measurement of d

$$d_1 = 0.800\text{ m},\ d_2 = 0.800\text{ cm}$$

$$\langle d \rangle = \tfrac{1}{2}(0.800 + 0.800) = 0.800\text{ m}$$

d = 0.800 m [1]

(iii) Estimate the percentage uncertainty in your value of d.

Considering thickness of the strings,

$$\Delta d = 0.002\text{ m}$$

Percentage uncertainty

$$\frac{\Delta d}{d} \times 100\% = \frac{0.002}{0.800} \times 100\% = 0.25\%$$

percentage uncertainty in d = 0.50% [1]

(iii) The distance between the bottom face of the top rule and the top face of the bottom rule is L, as shown in Fig. 1.

Measure and record L.

Measurement of L

$$L_1 = 0.304 \text{ m},\ L_2 = 0.306 \text{ m}$$

$$\langle L \rangle = \tfrac{1}{2}(0.304 + 0.306) = 0.305 \text{ m}$$

L = 0.305 m [1]

(iv) Estimate the percentage uncertainty in your value of L.

Uncertainty in L

$$\Delta L = 0.003 \text{ m}$$

Percentage uncertainty

$$\frac{\Delta L}{L} \times 100\% = \frac{0.003}{0.305} \times 100\% = 0.98\%$$

Note:

$\Delta l > \Delta d$.

The measurement of L is inherently imprecise due to

- different length of string
- the need to hold the measuring ruler still and vertically

The measurement of d is more precise as the strings do not move and it is resting directly on the scale.

Most candidates took repeated readings for l, but most judged the absolute uncertainties for l and d to be the same without considering the difficulties inherent in each particular measurement.

percentage uncertainty in L = 0.98% [1]

(b) **(i)** Give the suspended metre rule a small angular displacement about the vertical axis through the centre of the rule. Release the rule to set it into an angular oscillation, as shown in Fig. 2.

(ii) Record the time taken t for an appropriate number of oscillations N and calculate the period of oscillation T. State the number of oscillations clearly.

Number of oscillations

$$N = 30$$

Timing for 30 oscillations

$$t_1 = 23.44 \text{ s}, \; t_2 = 23.36 \text{ s}$$

$$\langle t \rangle = \tfrac{1}{2}(23.44 + 23.36) = 23.40 \text{ s}$$

Period

$$T = \frac{\langle t \rangle}{N} = \frac{23.40}{30} = 0.7800 \text{ s} \quad \text{(4 s.f.)}$$

Note:

t_1 and t_2 are recorded to nearest 0.01 s.

The measurement of t must be repeated.

The timing t must be greater than 20 s.

Number of s.f. for T is the same as t_1 and t_2.

t = 23.40 s ....

T = 0.7800 s .... [2]

(c) Repeat steps **(a)(ii)** and **(b)** to obtain 5 further sets of readings for *d* and the corresponding period *T* of the oscillations.

Always ensure that the two strings are equidistant from the ends of the suspended rule and vertical.

Present your results clearly. [5]

d / m	N	Time for N oscillations / s			T / s	$\frac{1}{d}$ / m^{-1}
		t_1	t_2	$\langle t \rangle$		
0.800	30	23.44	23.36	23.40	0.7800	1.25
0.700	25	22.16	22.10	22.13	0.8852	1.43
0.600	20	20.66	20.56	20.61	1.031	1.67
0.500	20	24.88	24.68	24.78	1.239	2.00
0.400	15	23.18	23.05	23.12	1.541	2.50
0.300	12	24.50	24.57	24.54	2.045	3.33

(d) Theory suggests that *T* and *d* are related by the expression.

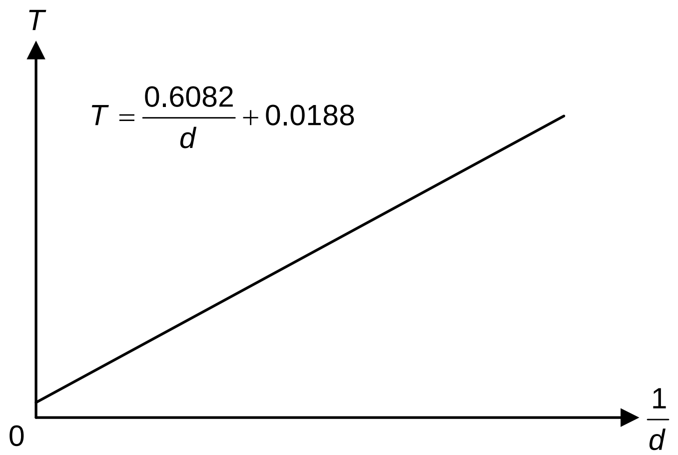
$$T = \frac{A\sqrt{L}}{d} + B$$

where *A* and *B* are constants.

(i) Plot a suitable graph using a spreadsheet to determine *A* and B.

Sketch the graph in the space below. Write down the equation of the trendline.

Plot a graph of T against $1/d$ where $A\sqrt{L}$ is the gradient and B is the vertical intercept.



$$A = \frac{0.6082}{\sqrt{0.305}} = 1.101 = 1.10 \text{ s m}^{0.5}$$

A = 1.10 s m$^{0.5}$ ....

B = 0.0188 s .... [4]

(ii) The constant A is related to the acceleration of free fall g by the following equation:

$$A = \frac{2\pi L_0}{\sqrt{3g}}$$

where L_0 is the length of the metre ruler.

Determine the value of g.

Length of metre ruler

$$L_0 = 1.000 \text{ m}$$

Acceleration of free fall

$$g = \frac{4\pi^2 L_0^2}{3A^2} = \frac{4\pi^2 \times 1.000^2}{3 \times 1.101^2} = 10.86 = 10.9 \text{ m s}^{-2}$$

g = 10.9 m s^{-2} [1]

(e) **(i)** State one significant source of error in this experiment.

The centre of mass of the suspended rule is moving as it oscillates. This introduces other modes of oscillation (e.g., left-right or front-back motion) and this may affect the accuracy of *T*.

The axis of rotation may not be at its centre of gravity. This may affect the accuracy of *T*. [1]

(ii) Suggest one improvement that could be made to the experiment to address the source of error identified in **e(i).** You may suggest the use of other apparatus or a different procedure.

Place a marker above/below the 50-cm mark of the suspended metre rule. When oscillating the ruler in the horizontal plane, ensure that the 50-cm mark remains above/below the marker so that other modes of oscillation will not be introduced.

Use a pointer to determine the position of the centre of gravity, where the axis of rotation will pass through. [1]

[Total: 18]

End of Paper